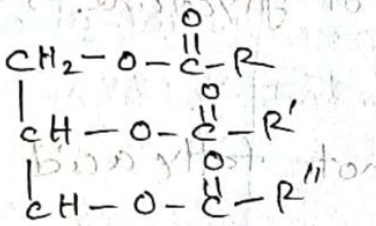
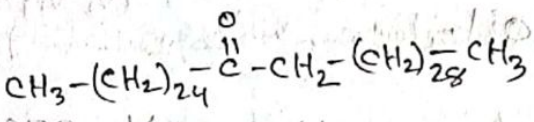
Lipids and Steroids

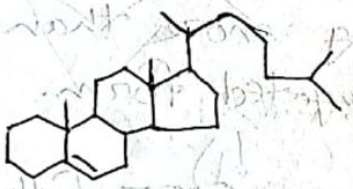
Lipid: Naturally occuring organic compounds which are isolated from plant or animal, soluble in organic solvent (such as ether, chloroform, benzene etc.), insoluble in polar solvents are called lipid. Example: Triglyceride



Triglyceride



Bees wax



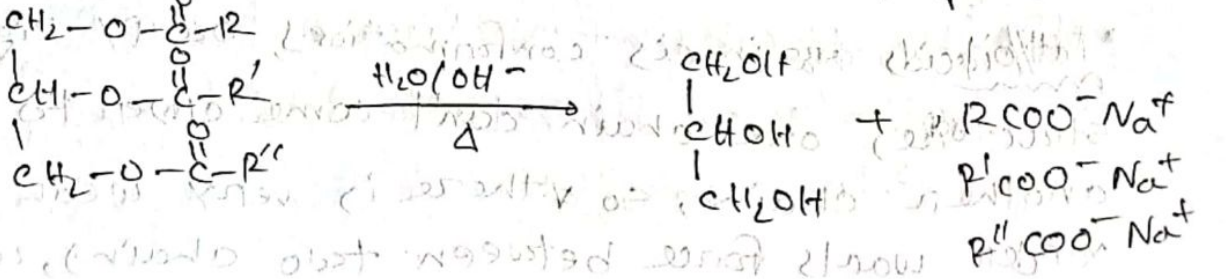
Cholesterol

Classification:

① Hydrolysable (Triglyceride, oil and Fat)

② Non-hydrolysable (cholesterol, vitamin B., Vit. A)

→ Lipids containg carboxylic acid derivative functional groups are generally hydrolysable. Hydrolysable lipids contains ester, amide etc. functional group.

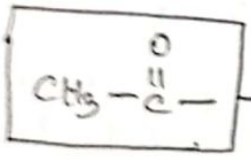


Fatty acid: Fatty acids are long chain carboxylic acids obtained by hydrolysis of lipid. Fatty acids occur naturally, possess carbon atoms in even number and are usually unbranched. Example: stearic acid.

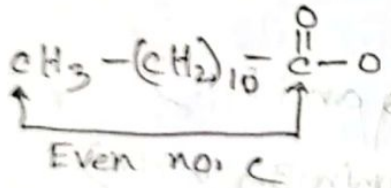
Fat/oil: The long chain triesters of glycerine is called fat or oil.

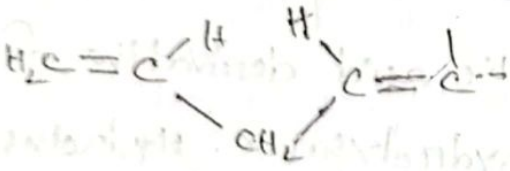
Note: All carboxylic acids are not fatty acid

→ Fatty acids always have long chain, without branching.
→ There are even no of carbon carbon atom.
→ unsaturated fatty acids having more than one double bond are never in conjugated form.



repeatedly এই unit থাকলে তাকে fatty acid বলি হয়।



→ 

cis form

Fat/oil have cis conformations, occupy zigzag structure, one chain can't come closer to another chain, so there is very weak vander waals force between two chains, which

make the oil liquid in room temp.

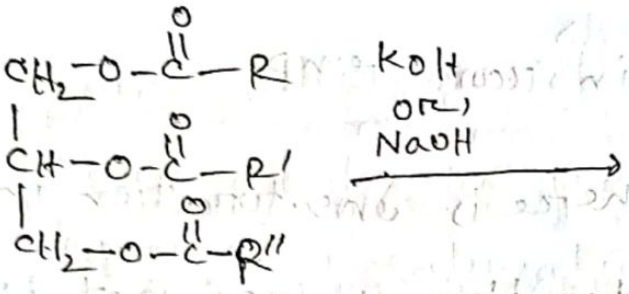
on the other hand, there is unsaturation in fatts/oils, but the double bonds are not in conjugated form, there is a active methylene group ($-CH_2-$) present between two double bond.

$$CH_3-(CH_2)_n-CH=CH-\underset{\text{active methylene group.}}{\overset{\uparrow}{CH_2}}-CH=CH-$$

The active methylene group is responsible for cross-linking between chains. [Lower the unsaturation, less zigzag it is, chain can come closer, higher the cross linking between chain, which solidify the fats. So fats having lower unsaturation are solid in room temp.] (Lower unsaturation, less zigzag, come to closer, good cross linking, solid)

Q. why fat is solid, oil is liq.?

Saponification value: The no. of miligrams of sodium or potassium hydroxide required to saponify one gram of fat under the condition specified. It is the measure of the average molecular weight of all the fatty acids present in the sample. in form of triglycerides. (since R groups are not same in every molecule of oil/fat).



$$\therefore \text{Saponification value} = \frac{(V_2 - V_1) \times N \times M}{w}$$

w = wt. of oil/fat
V_2 = Blank test
V_1 = NaOH consumed in actual test
N = Normality of NaOH
M = Molar weight of NaOH.

Simply, Saponification value

$$S = \frac{\text{Mass of NaOH req. (in mg)}}{\text{wt. of fat/oil (in g)}}$$

Now, assuming two oils A and B having saponification value 50 and 70

$S_A \rightarrow 50$ mg/g
$S_B \rightarrow 70$ mg/g

That means 50 mg NaOH req. to completely saponify 1 g of oil A, and 70 mg req. for 1 gm of oil B.

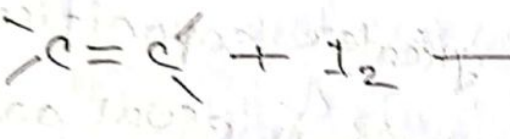
Case: 1
→ May be R chain is larger for oil A compared to oil B. Larger the R chain higher the mass, lower the quantity in

1 mg sample. Hence, less amount of NaOH/KOH required to saponify it, means lower saponification value.

Case 2: we can get greater saponification value event if the chain is equal or longer than the other. Assuming for a moment that Oil B has larger R chain, but highly unsaturated R chain. Due to unsaturation there is less Hydrogen present in the chain, the overall mass of the molecule reduces due to unsaturation. and quantity in 1 mg sample increases, to saponify the larger quantity, large amount of KOH or NaOH required. So saponification value is higher.

Decision: Oil A have higher M.W. due to larger R chain or unsaturation in the alkyl chain (12). That why it has lower saponification value than A.

Iodine value: No. of grams of iodine absorbed by 100 g of fat. It indicates the degree of unsaturation of a fat or oil.



do not

→ usually iodine undergo reaction with double bond, so we use inter halogen compound I-Cl in lab experiment. Inter halogen compounds are highly reactive.

No of iodine value ∝ double bond, unsaturation

Data:

② Iodine value of,

soya = 127-138

coconut oil = 08

① Saponification value of,

soya bean = 180-200

coconut oil = 250-268

* From the given data justify which oil is healthy for us.

Ans: Unsaturated oils helps to lower cholesterol and improve heart health.

From the definition of saponification value (সংজ্ঞা definition থেকে পাওয়া),

coconut oil can have higher saponification value, due to small R chain or unsaturation.

But the iodine value shows coconut oil have less unsaturation than soyabean oil.

That means coconut oil must have small alkyl chain, with lower unsaturation.

But soyabean oil has greater unsaturation (since iodine value is very high).

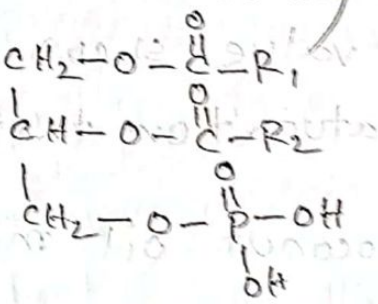
So soyabean oil is healthy than coconut oil.

Note:

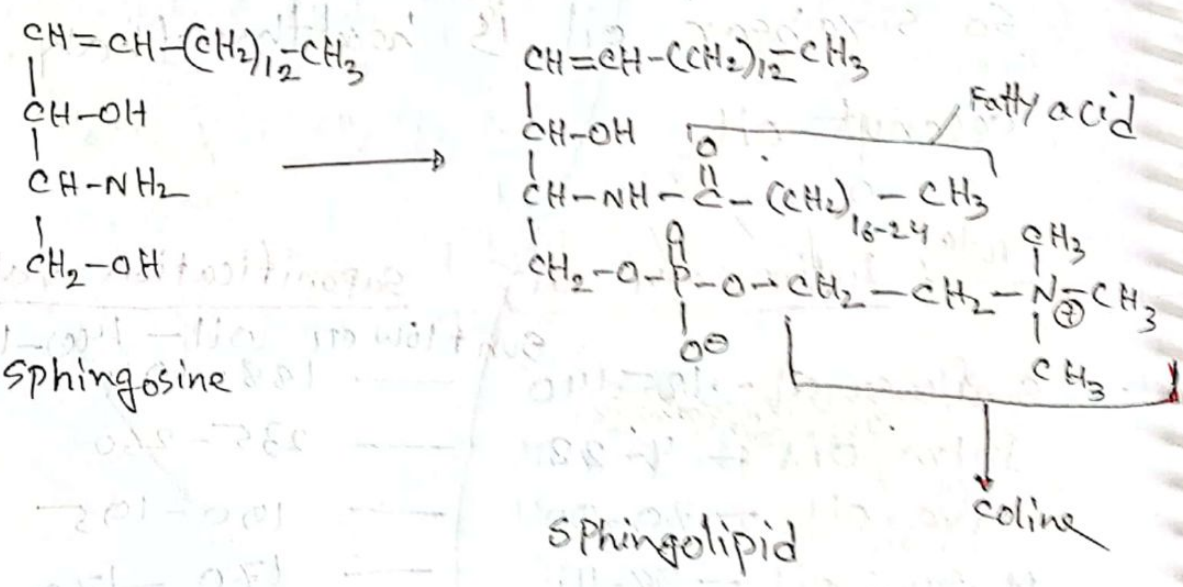
	Iodine value of,	Saponification value
Sunflower oil	125-140	188-194
Palm oil	4-22	235-260
Olive oil	79-90	190-195
Mustard oil	98-110	170-178

CamScanner

Phospholipids: A phospholipid is composed of a glycerol backbone attached on one end to two fatty acids and the other end has the esterified phosphoric acid and organic alcoholic group. In other words, (a lipid containing a phosphate group in its molecule.)



Sphingo lipid: A class of lipid containing a backbone of sphingoid bases, a set of aliphatic amino alcohols that includes sphingosine.



Sphingosine

Sphingolipid

→ Phospholipid obtained from glycerine backbone is called phosphoglycerides.

→ Phospholipid obtained from sphingsine are called ~~Sphingolipid~~/Sphingomyelin.

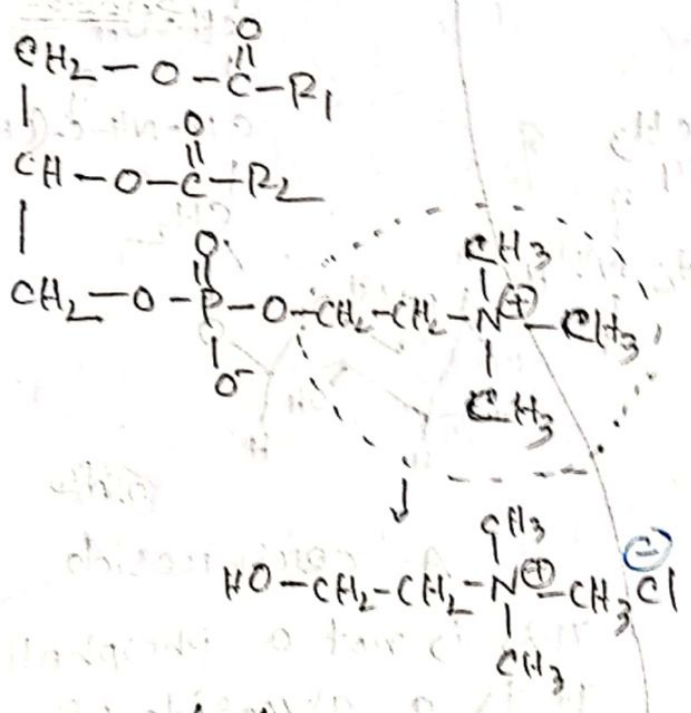
According to structural backbone Phospholipids are two types:

① Phosphoglyceride

② Sphingomyelin.

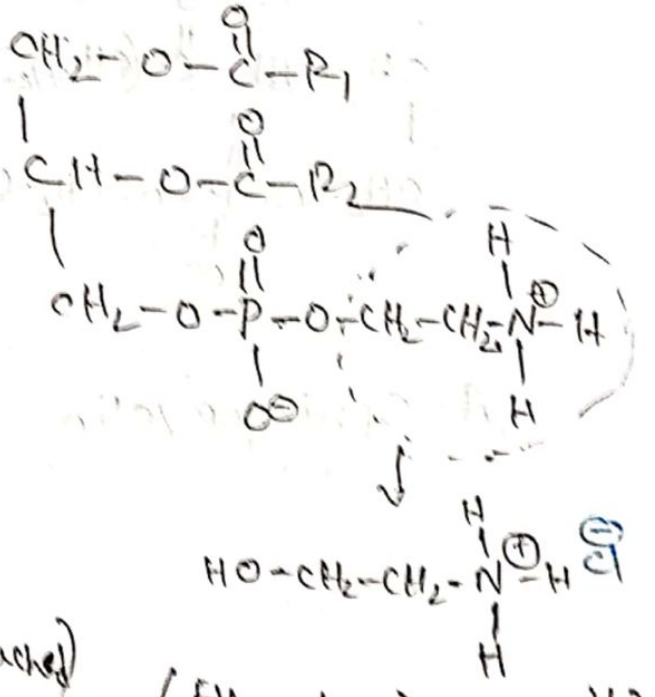
* Phosphoglyceride 2 types:

① Lecithene



(choline chloride attached)

② cephalin



(Ethanolamine chloride)

* Sphingo lipid 3 types (according to attached group)

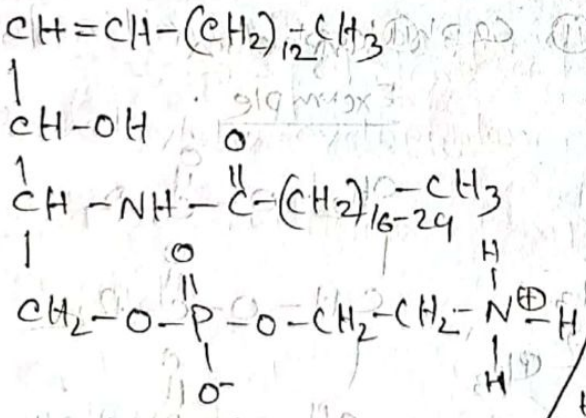
① choline chloride
② Ethanol amine chloride } Type 1
③ carbohydrate } Type 2

Actually 2 type

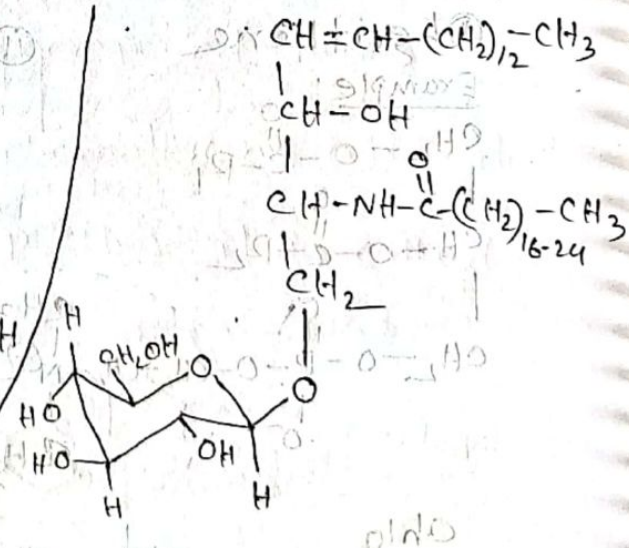
Type 1 - Sphingomyelin

Type 2 - Cerebroside

Example



A Sphingomyelin

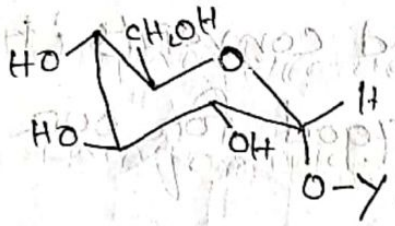


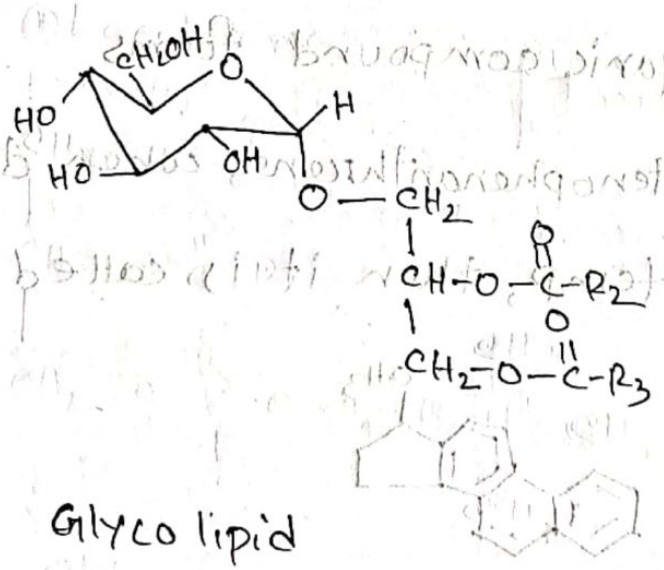
A cerebroside

This is not a phospholipid, It is a glycoside (forms glycosidic bond with anomeric c)

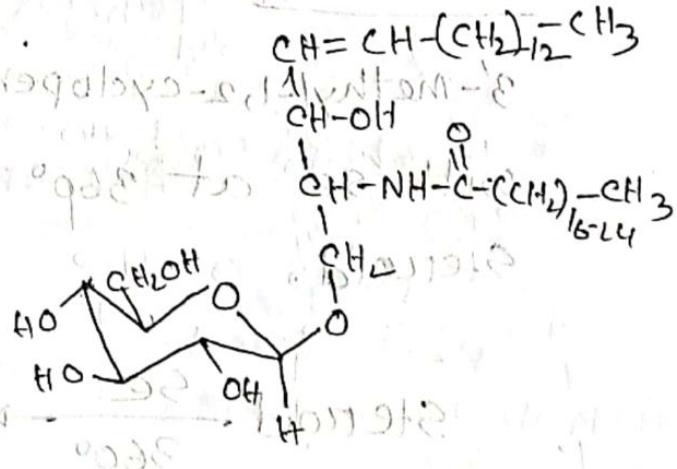
* All sphingolipids are not phospholipid.
* All sphingomyelin are phospholipid, but cerebrosides are not.

Glycolipids: Glycolipids are lipids with a carbohydrate attached by a glycosidic bond





Glyco lipid



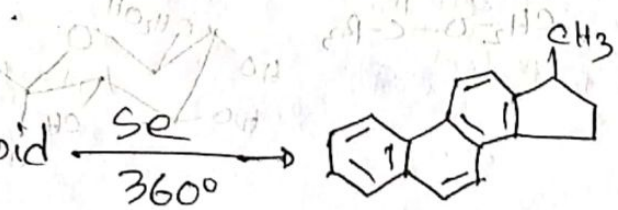
Sphingolipid

HDL – High density lipoprotein (40% protein, 60% lipid)
LDL – Low density lipoprotein (25% protein, 75% lipid)

→ cholesterol is biosynthesized in our body. LDL bring cholesterol from our liver to cell of our body and accumulate it on the cell surface. while HDL carry the cholesterol from cell surface to the liver and converts it in to binacid. so HDL is good for our health

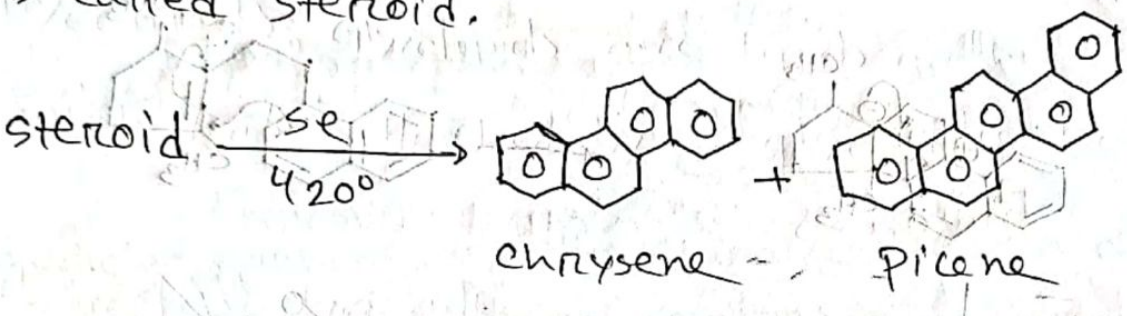
Part B: steroids

Definition: If any organic compound forms 3'-methyl-1,2-cyclopentenophenanthrene when distilled with Se at 360° temp, then it is called steroid.

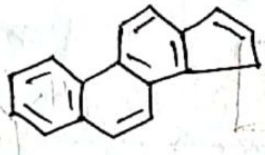


Diels hydrocarbon/
3'-methyl-1,2-cyclopentenophenanthrene

Another definition: If any organic compound forms chrysene and picene when distilled in presence of se at 420°c, then the compound is called steroid.



* 1,2-cyclopentenophenanthrene is the structural unit of steroid.



Structure determination of Diels hydrocarbon

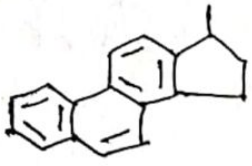
1) M.w is determined doing elemental analysis.

2) DBE = 19 − 8 = 11

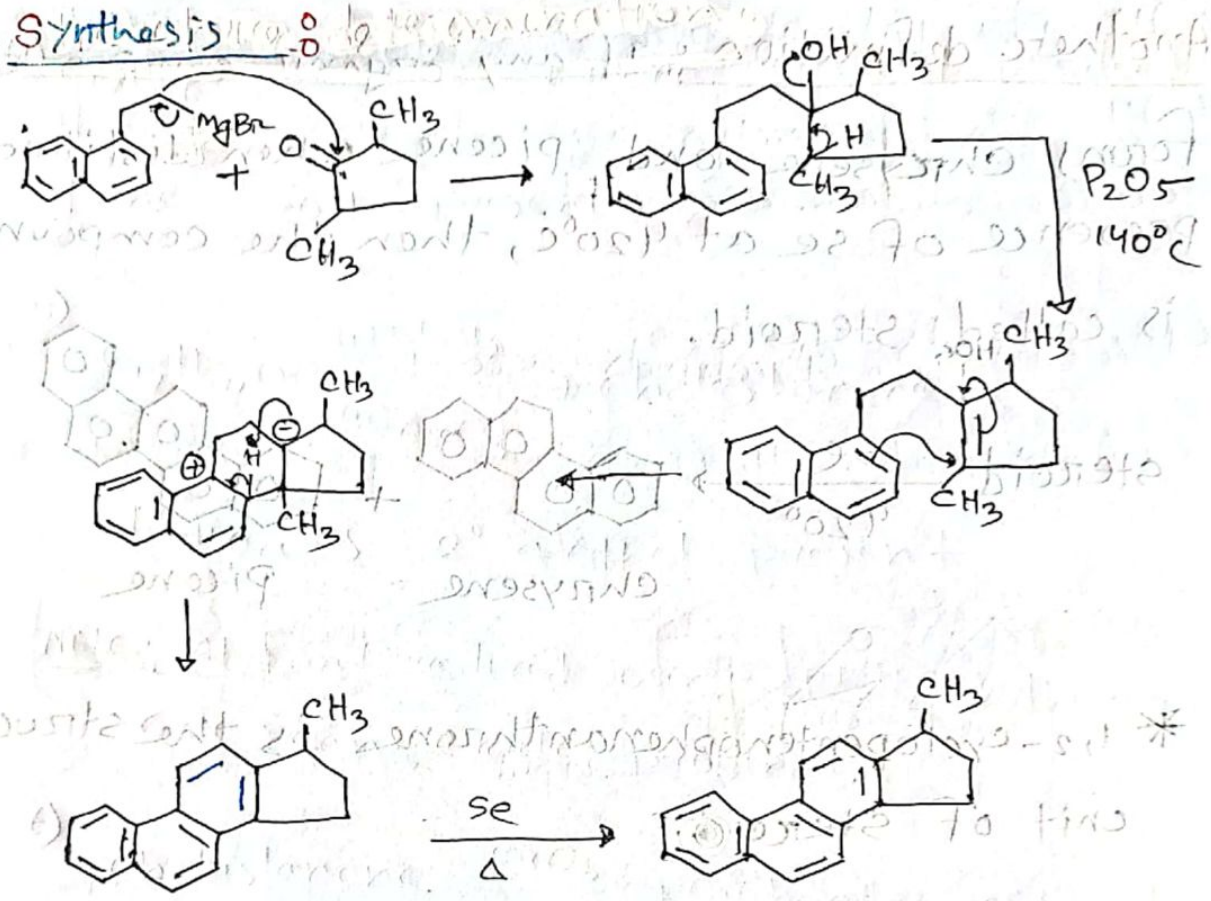
3) $C_{18}H_{16} \xrightarrow[Ni/Pd/Pt]{7\ H_2} C_{18}H_{30}$

Ring = 11 − unsaturation = 11 − 7 = 4

4) Melting point = 126 − 127°c

5) x-Ray crystalography show) → 

Synthesis



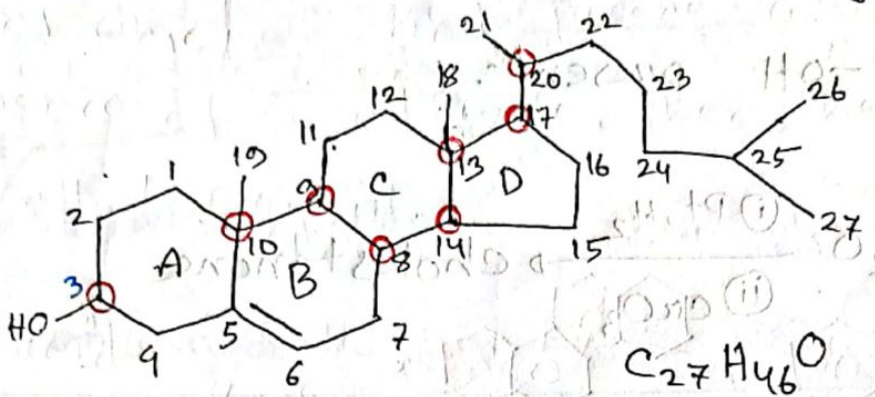
Cholesterol

* Is cholesterol a steroid? How can you establish it?

cholesterol $\xrightarrow[360°C]{Se}$ 3'-methyl-1,2-cyclo

So cholesterol is a steroid.

* Draw the structure of cholesterol numbering all the carbon atoms and showing chiral centres.



$C_{27}H_{46}O$

→ Circled carbons are the chiral carbon in cholesterol. There are eight chiral centres.

→ cholesterol has eight chiral centres, so $2^8 = 256$ sterioisomer is possible, of which only one occurs in nature

Structure elucidation of cholesterol:

1) From elemental analysis, M.F = $C_{27}H_{46}O$

2) DBEV = $C + 1 - \frac{H}{2} = 5$

3) (i) $C_{27}H_{46}O \xrightarrow[Pt/Ni/Pd]{H_2} C_{27}H_{48}O$

2-H added means one C=C or C=O

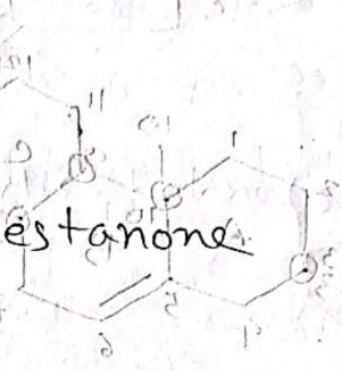
(ii) $C_{27}H_{46}O \xrightarrow{Br_2} C_{27}H_{46}OBr_2$

Indicates one C=C present.

4) $C_{17}H_{46}O \xrightarrow{(CH_3CO)_2O}$ mono acetilated product

one $-OH$ present.

5) $C_{17}H_{46}O \xrightarrow[\text{(ii) } CrO_3]{\text{(i) } Pt, H_2}$ cholestanone



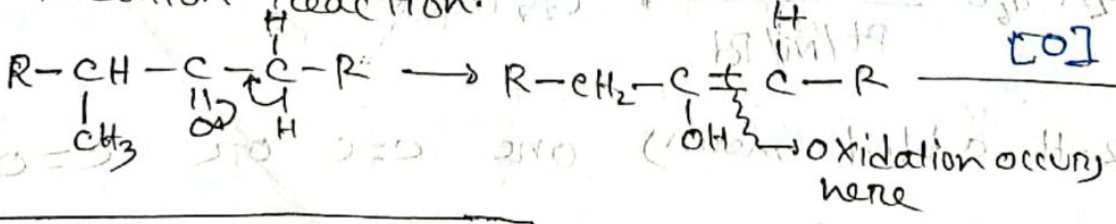
Indicates 2° $-OH$ present

Note: At first reduced using Pt, H_2, otherwise $C=C$ can create problems.

6) Cholestanone $\xrightarrow{CrO_3}$ Carboxilic acid without losing carbon

Indicates the $-OH$ group attached to the ring.

Explaination: ketones do not participate in oxidation reaction directly. If there is a proton present at α position, then it occupy enolic form and undergo oxidation reaction.

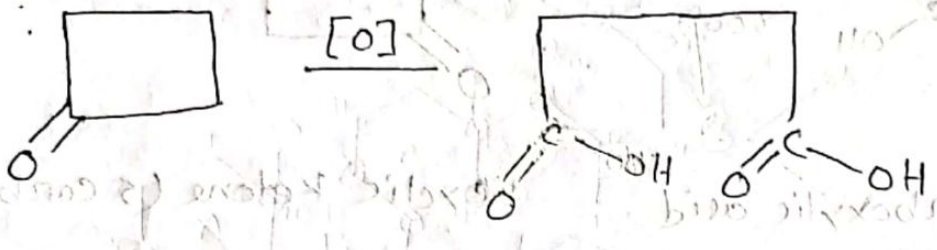


2 Products, means no product contains equal no of carbon present in the reactant.



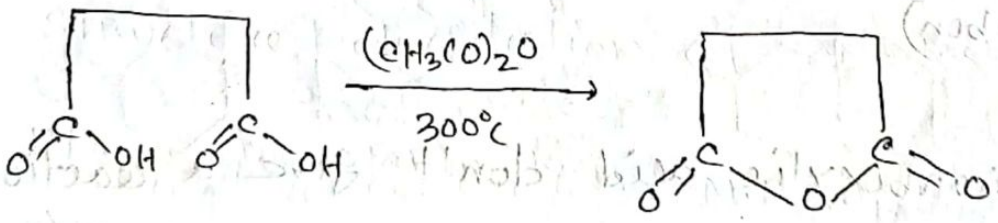
Mixture of carboxylic acid

But in case of cyclic ketone

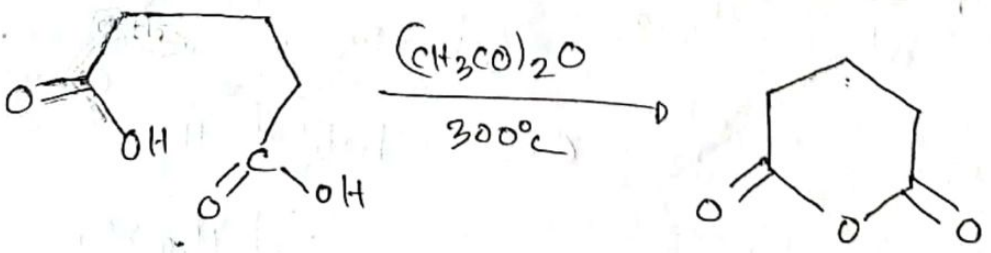


No. of carbon remains same.

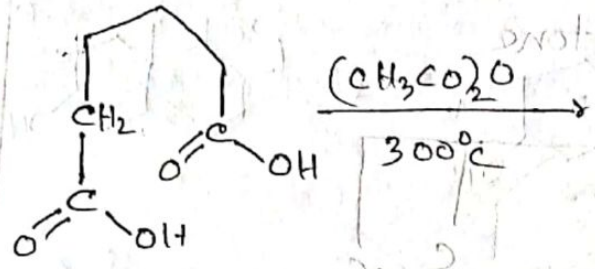
Blanc's rule: 1,4 and 1,5- dicarboxilic acids gives anhydrides while 1,6 and 1,7- dicarboxylic acid give ketones with elemination of CO_2 on heating at 300°C. This is called Blanc's rule.



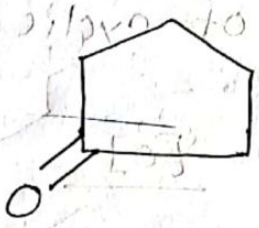
1,4 dicarboxilic acid — cyclic Anhydride



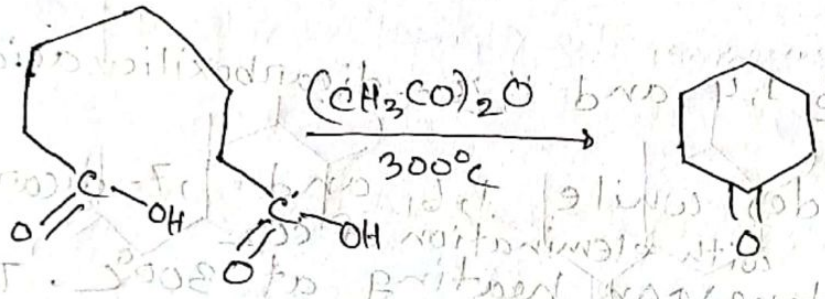
1,5-dicarboxylic acid — cyclic Anhydride



1,6-dicarboxylic acid — 6 carbon



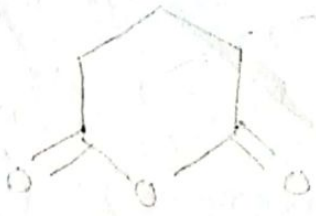
cyclic ketone (5 carbon)

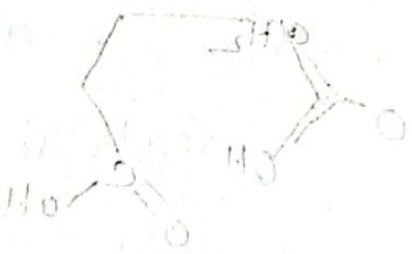


1,7-dicarboxylic acid (7 carbon)

cyclic ketone (6 carbon)

→ If dicarboxylic acid don't give reaction then there way above 7 carbon or below 4 carbon present in reactant.





1,7-dicarboxylic acid $\xrightarrow[300^\circ]{(CH_3O)_2O}$ 6 membered cyclic ketone

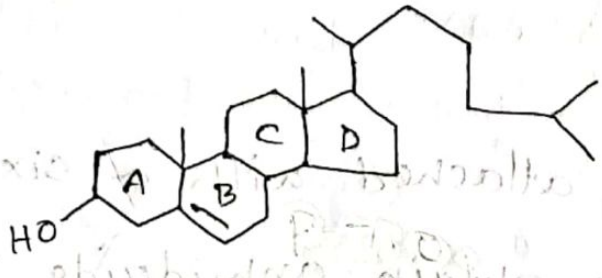
6 membered cyclic ketone $\xrightarrow{[O]}$ 1,6 diconboxylic acid

1,6 diconboxylic acid $\xrightarrow[(CH_3O)_2O]{300^\circ C}$ 5 membered cyclic ketone

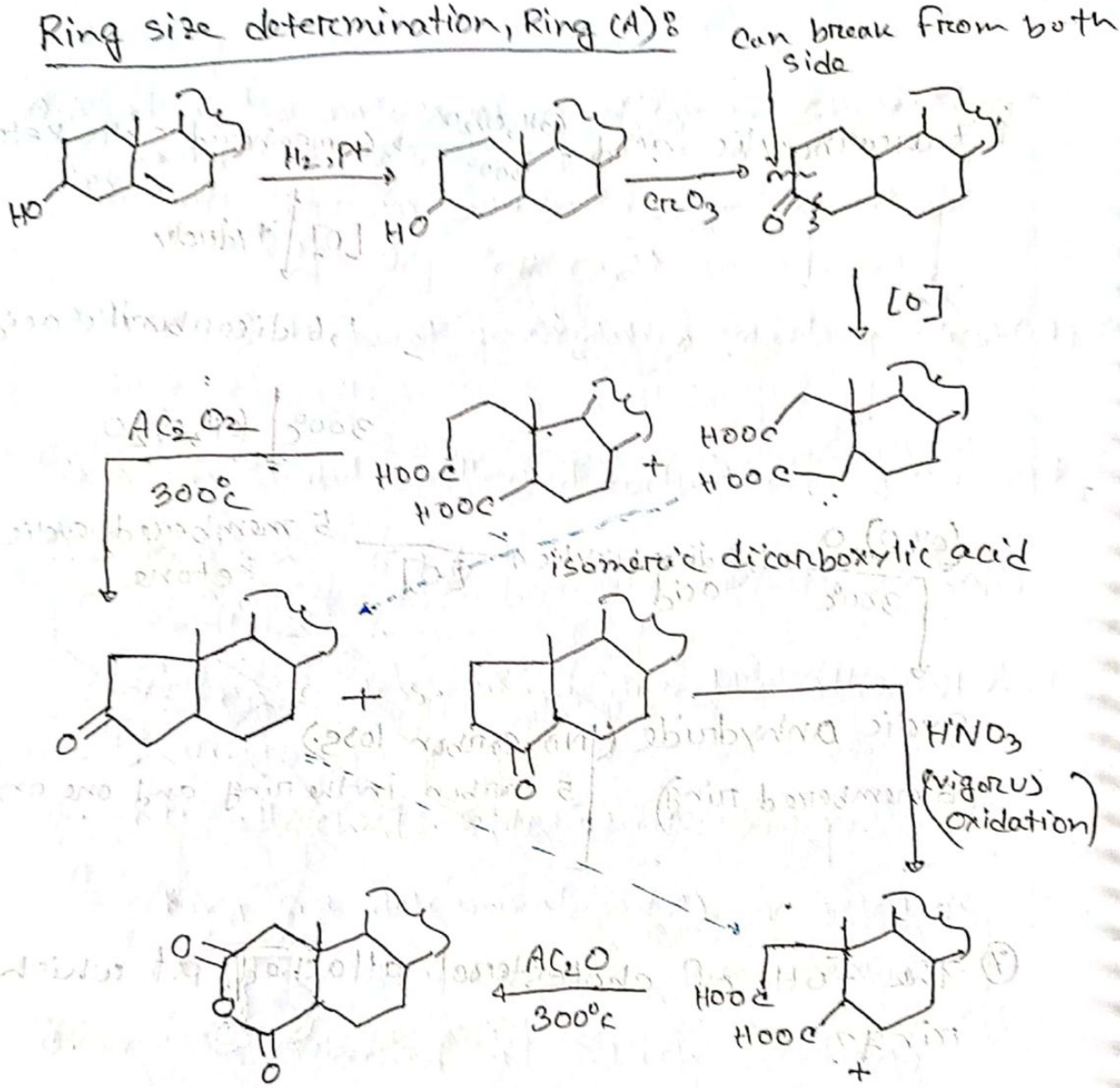
5 membered cyclic ketone $\xrightarrow{[O]}$ 1,5-dicarboxylic acid $\xrightarrow[300^\circ C]{(CH_3O)_2O}$ cyclic anhydride (6 membered ring)

[No carbon lose
5 carbon in the ring and one oxygen]

⑦ The -OH of cholesterol attached at which ring?

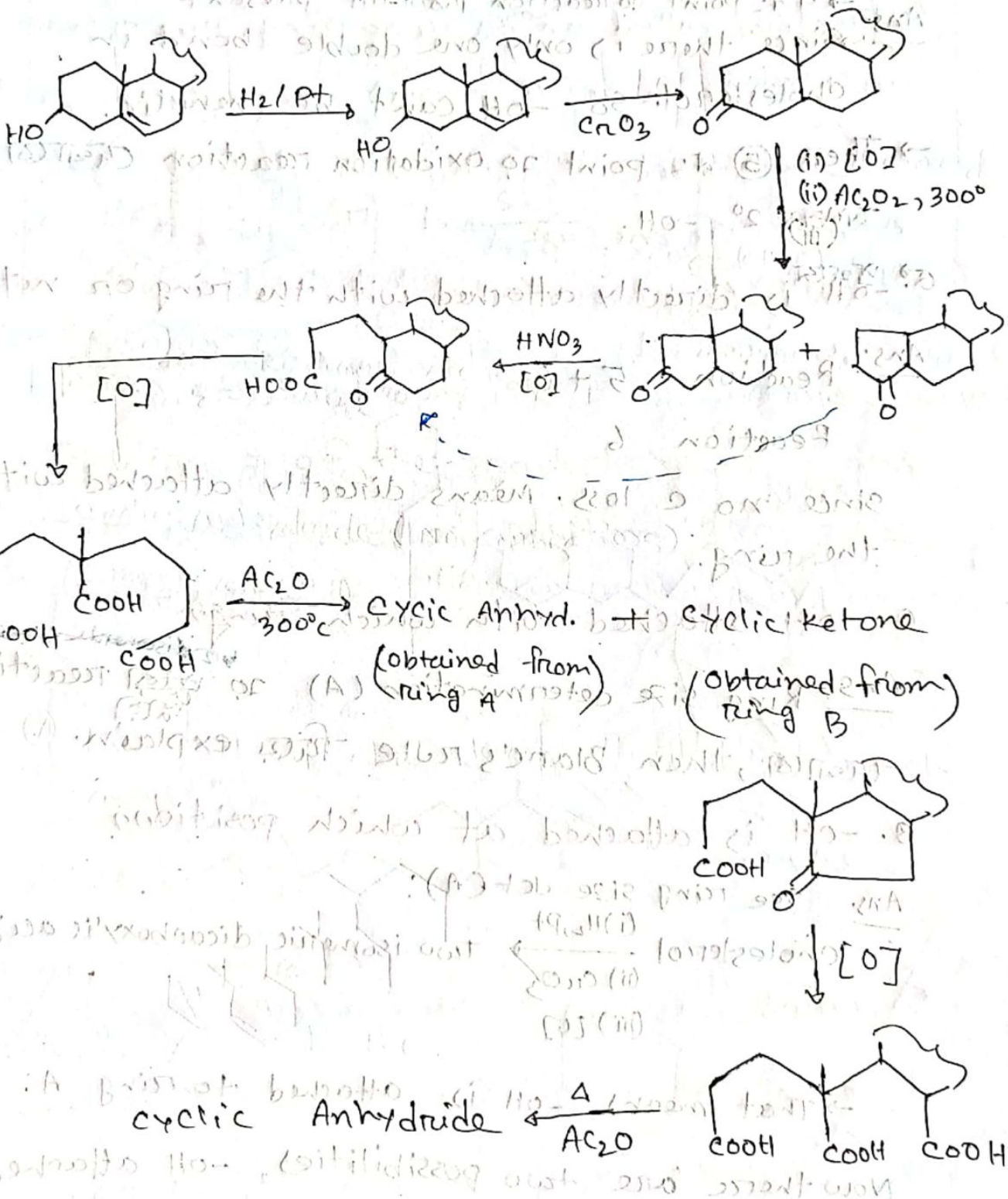


Ring size determination, Ring (A):



→ -OH must be attached with a six membred ring, so that we obtain anhydride in this step. If the ring was seven membered we would obtain cyclic ketone, but we obtain anhydride.

Ring size (B) determination



Indicates the ring B is 6 membered

Q. what is the nature of O atom in cholesterol?

Ans: → 4 নং point এ reaction দিয়ে -OH present

Since there is only one double bond in cholesterol. So -OH can't be phenolic.

→ Then ⑤ নং point এ oxidation reaction দেখানো, অর্থাৎ 2° -OH.

Q. -OH is directly attached with the ring or not?

Ans: Reaction 5

Reaction 6

since no C loss. means directly attached with the ring.

Q. -OH attached with which ring?

Ans: Ring size determination (A) এ দিয়ে reaction দেখানো, then Blanc's rule দিয়ে explain.

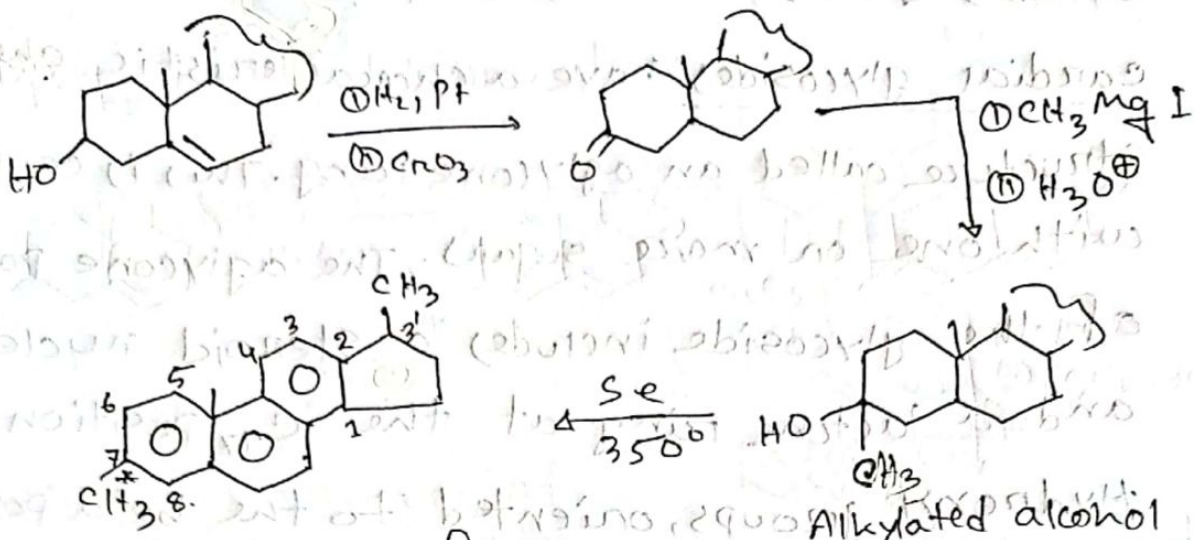
Q. -OH is attached at which position?

Ans. See ring size det (A):

Cholesterol $\xrightarrow[\text{(ii) } CrO_3 \text{ (iii) } [O]]{\text{(i) } H_2, Pt}$ Two isomeric dicarboxylic acid

→ That means -OH is attached to ring A.

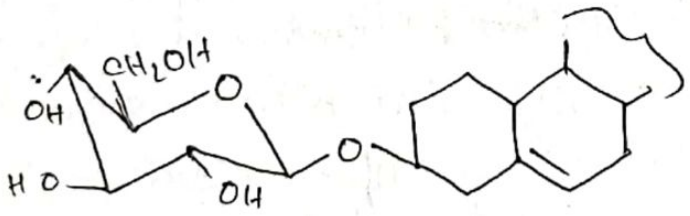
Now there are two possibilities, -OH attached either at position 2 or 3



Alkylated alcohol

$-CH_3$ at position 7 of diels hydrocarbon.

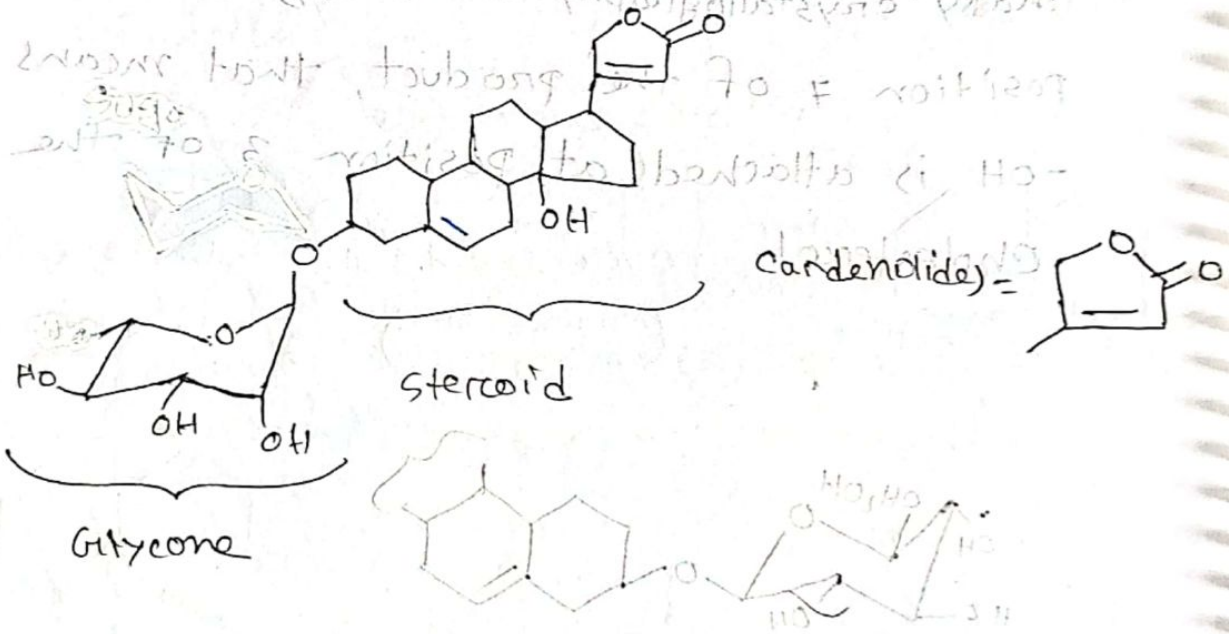
X-Ray crystallography shows alkylation occurs at position 7 of the product, that means -OH is attached at position 3 of the cholesterol.



Steroidal glycoside.

Syllabus এ আছে, but class এ পড়ানো হয়নি।

Cardiac glycosides have a characteristic chemical structure called an aglycone ring. This is coupled with one or more sugars. The aglycone portion of the glycoside includes a steroid nucleus and a lactone ring at the C_{17} position. Hydroxyl groups, oriented to the beta position are present at the C_3 and C_{14} positions.

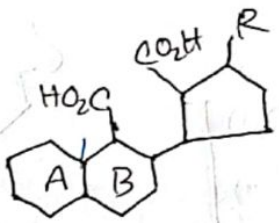


Saponins: Saponins contain a carbohydrate moiety attached to a triterpinoid or steroid. They can form stable, soap like foam in aq. soln.

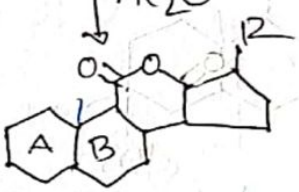
* one kind of "steroidal glycoside" forming foam in water (like soap) is known as saponins.

Ring size C:

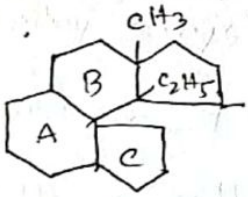
Cholesterol $\xrightarrow{\text{Deoxycholic acid}}$



$\downarrow Ac_2O$



Thus the structure of cholesterol should be

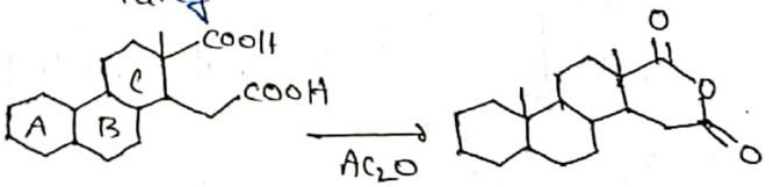


X-Ray crystallography confirms this is not the structure. So the ring C must be six membered.

Ring size-D

Cholesterol is converted into 5β-Cholestane, which was further converted into etiobilianic acid, this gave a cyclic anhydride. Hence, ring D is five membered.

* 5β-cholestane $\longrightarrow$



Etiobilianic acid

(1,5-dicarboxylic acid)